Experiment 4: Harris Corner Detection in Grayscale Images

Aim:

To manually compute and visualize Harris corner responses in a grayscale image.

Theory:

The Harris Corner Detection algorithm is used to identify points in an image where there is a significant change in intensity in all directions — typically corners or junctions. It works by computing image gradients (Ix, Iy), their products (Ixx, Iyy, Ixy), and then applying a mathematical formula to calculate a response value (R) for each pixel. High values of R indicate strong corners. These are then filtered using a threshold and visualized.

Algorithm:

• Start
• Import required libraries (`cv2`, `numpy`)
• Read input image and convert it to grayscale
• Compute image gradients using Sobel operator (Ix, Iy)
• Calculate Ixx, Iyy, and Ixy (squared gradients and product)
• Apply Gaussian blur to these matrices
• Compute Harris response using the formula: R = det(M) - k(trace(M))^2
• Apply threshold to extract strong corner points
• Mark corners on the image (in grayscale, RGB, and BGR versions)
• Display the results
• End

Flowchart:

Start → Import Image → Convert to Grayscale → Compute Ix and Iy → Calculate Ixx, Iyy, Ixy → Apply Gaussian Blur → Compute Harris R → Threshold → Mark Corners → Display Output → End

Viva Questions:

• What is the mathematical formula for Harris response?
• Why are Sobel derivatives used before Harris detection?
• How is thresholding used to identify corners?